

VAR - Overview

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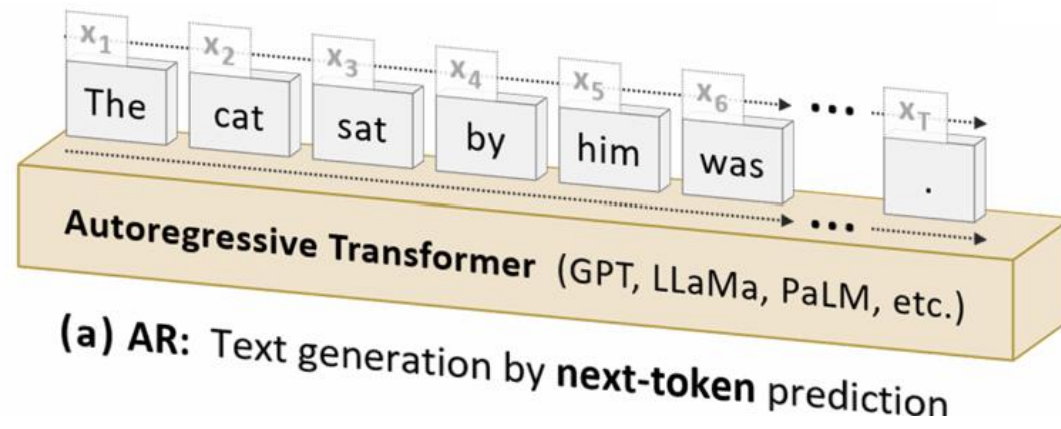
Autoregressive Modeling in Language Models

$$p(x_1, x_2, \dots, x_T) = \prod_{t=1}^T p(x_t \mid x_1, x_2, \dots, x_{t-1}).$$

$x = (x_1, x_2, \dots, x_T)$, a sequence of **discrete** tokens where $x_t \in [V]$,

known as '**next-token prediction**'

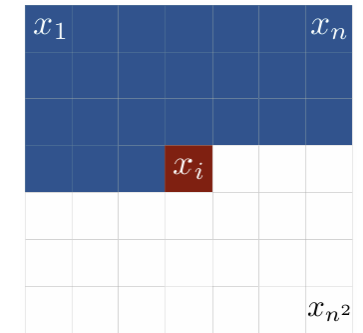
e.g. GPT, LLaMa, ...



Conventional Autoregressive Models in Vision

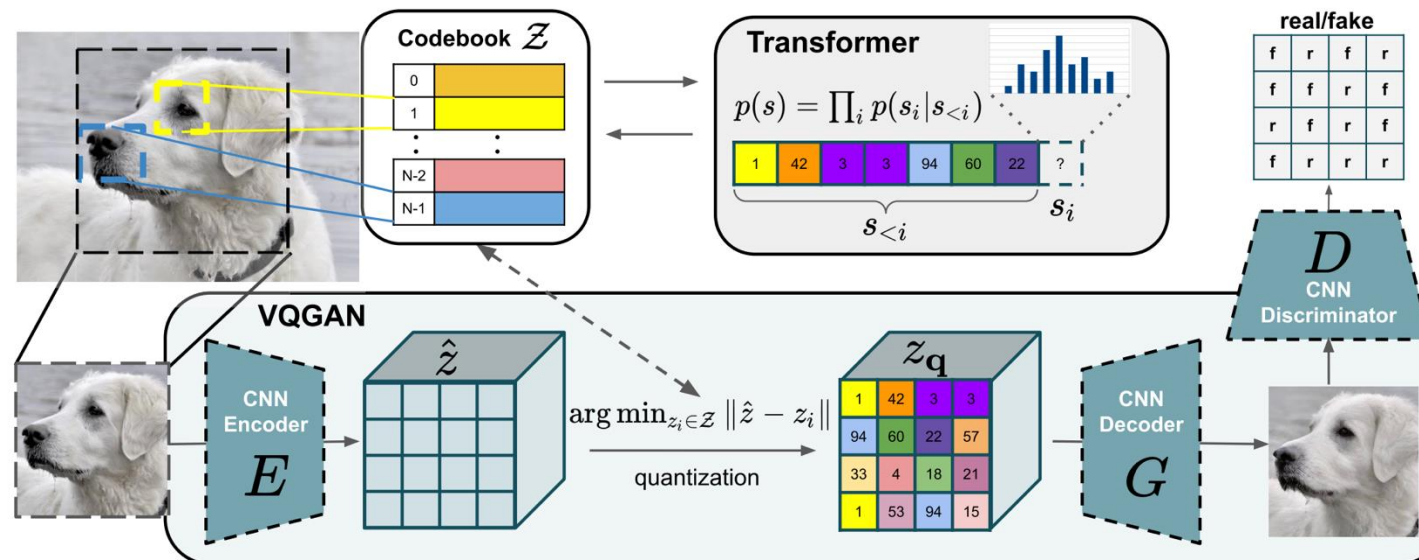
PixelCNN: Pixel-by-Pixel AR

VQGAN+Transformer: Latent token-by-token AR



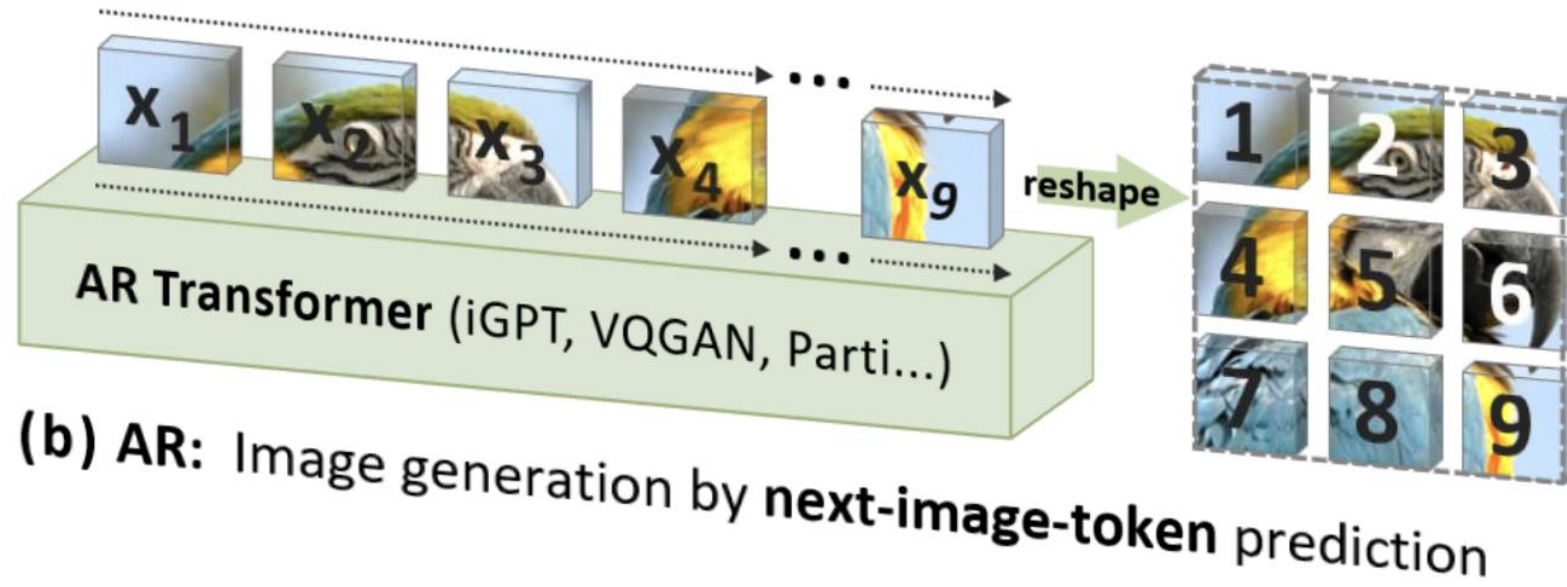
Context

$$p(\mathbf{x}) = \prod_{i=1}^{n^2} p(x_i | x_1, \dots, x_{i-1})$$

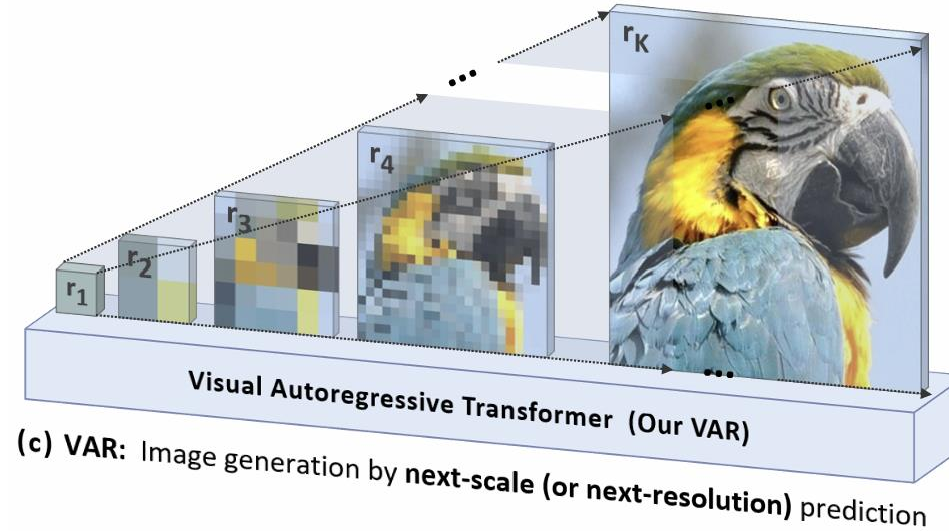


Problem of Conventional Vision AR

1. **Less competitive image quality** than diffusion models
2. **Slow** token-by-token generation



Visual AutoRegressive Model (VAR)

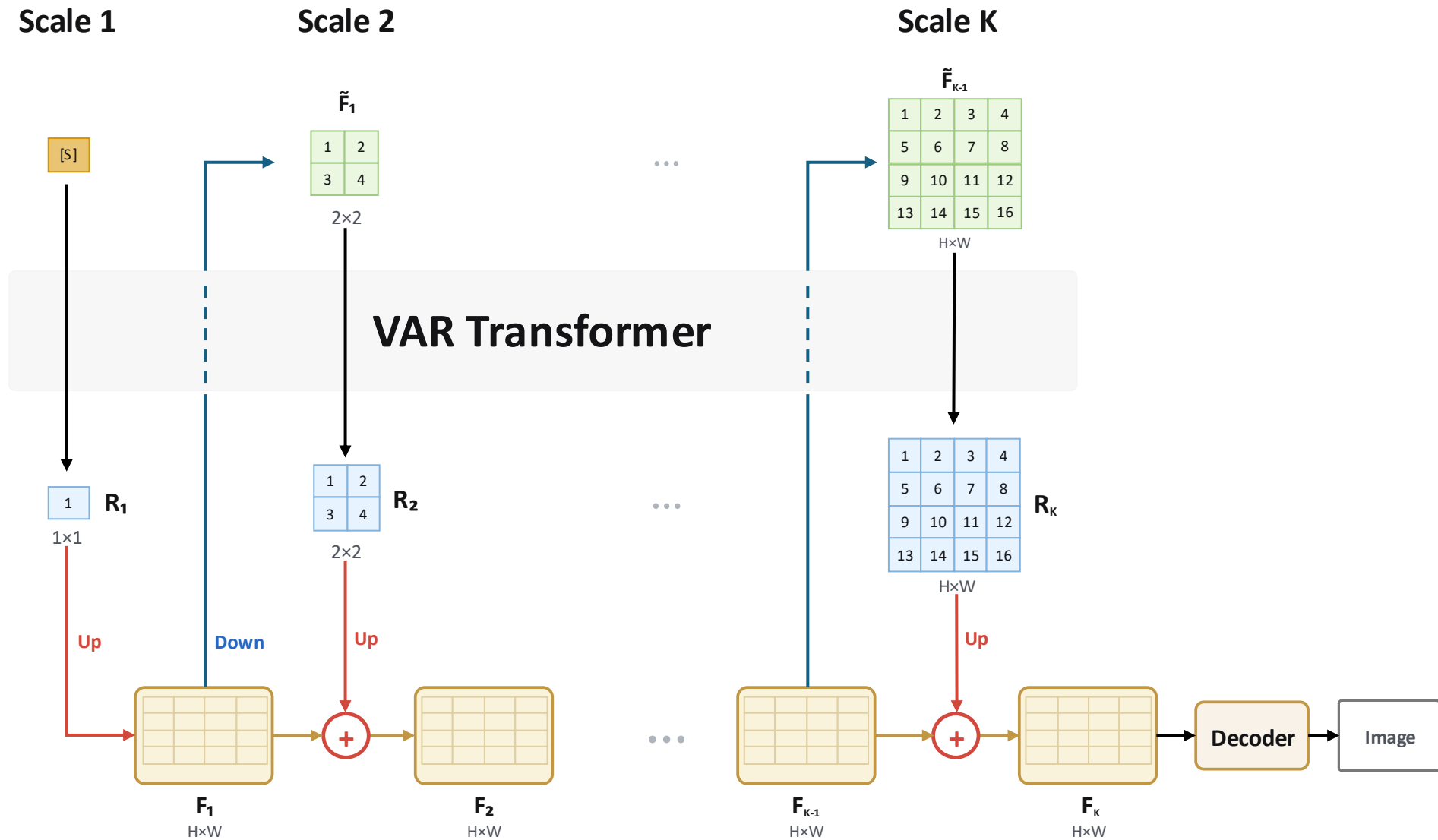


VAR reformulated the autoregressive modeling by **next-scale prediction**, instead of next-token prediction.

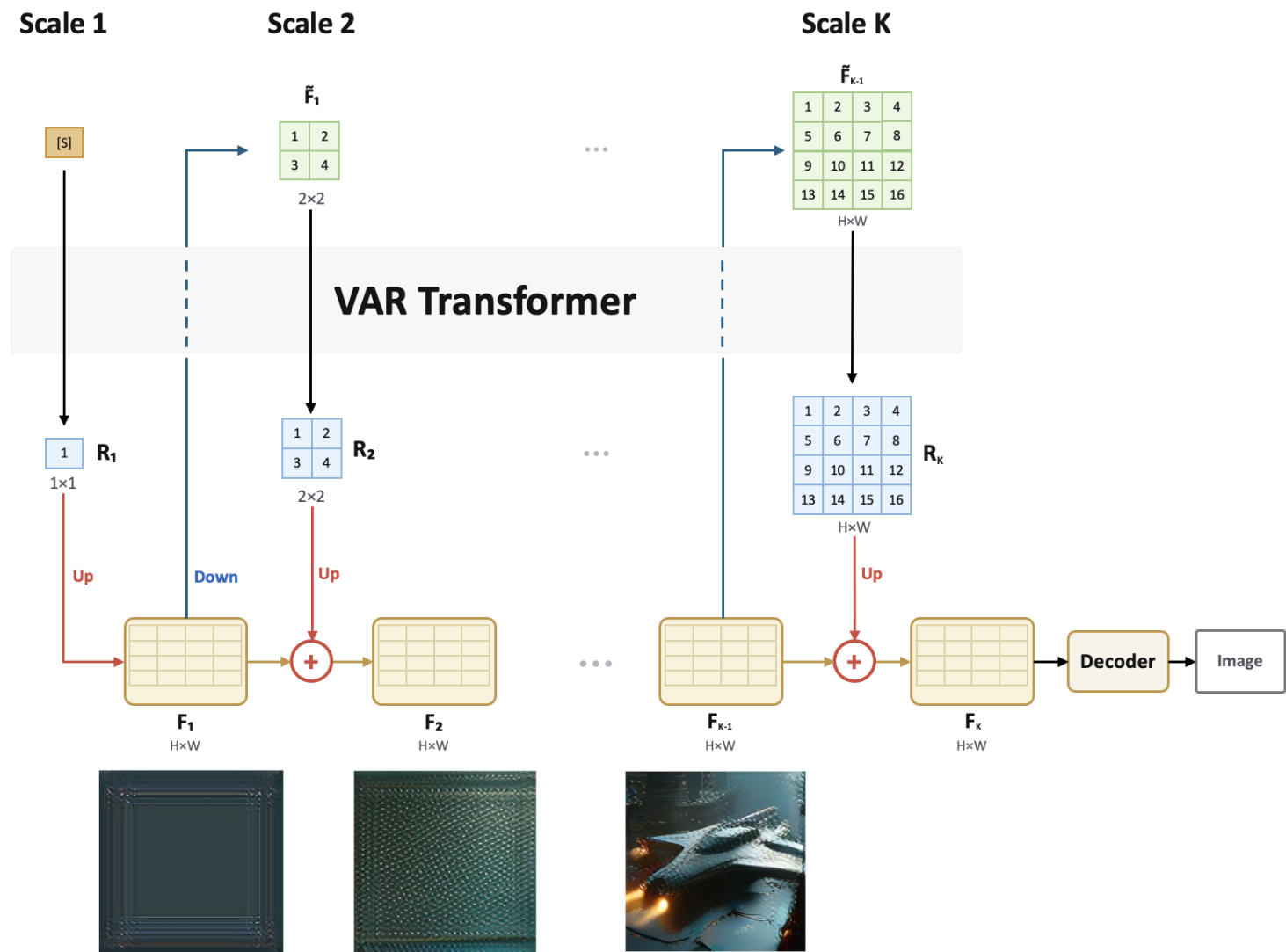
$$p(r_1, r_2, \dots, r_K) = \prod_{k=1}^K p(r_k \mid r_1, r_2, \dots, r_{k-1}),$$

$r_k \in [V]^{h_k \times w_k}$ is the token map at scale k containing $h_k \times w_k$ tokens

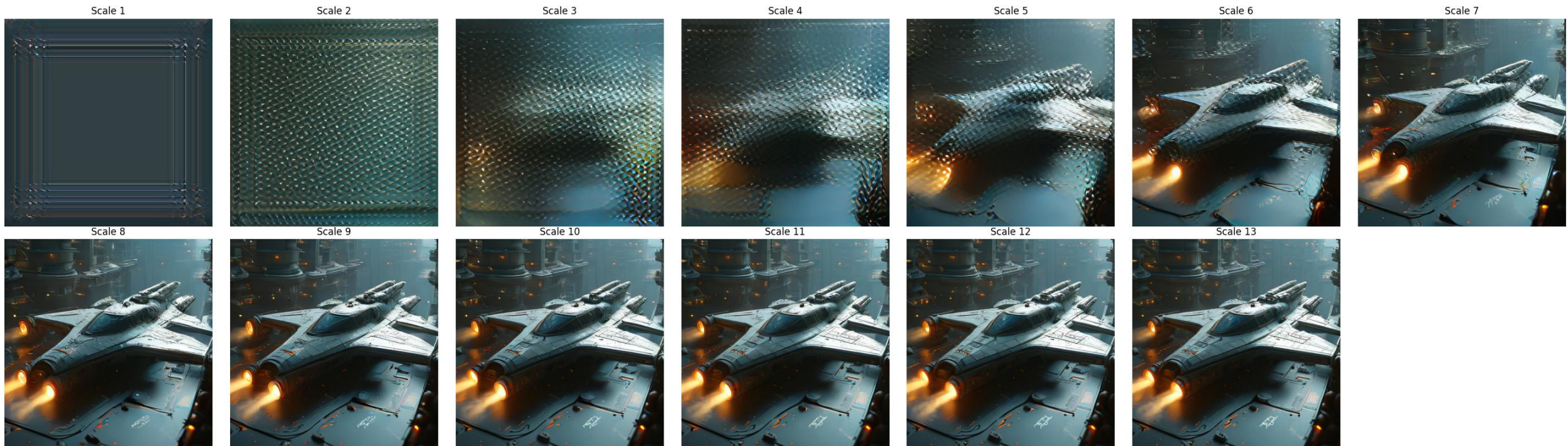
VAR Inference Mechanism



VAR Inference Mechanism



VAR Visualization by Scale



Result of Visual AutoRegressive Model (VAR)

Table 1: **Generative model family comparison on class-conditional ImageNet 256×256 .** “↓” or “↑” indicate lower or higher values are better. Metrics include Fréchet inception distance (FID), inception score (IS), precision (Pre) and recall (rec). “#Step”: the number of model runs needed to generate an image. Wall-clock inference time relative to VAR is reported. Models with the suffix “-re” used rejection sampling. †: taken from MaskGIT [17].

	Type	Model	FID↓	IS↑	Pre↑	Rec↑	#Para	#Step	Time
Diffusion	GAN	BigGAN [13]	6.95	224.5	0.89	0.38	112M	1	—
	GAN	GigaGAN [42]	3.45	225.5	0.84	0.61	569M	1	—
	GAN	StyleGan-XL [74]	2.30	265.1	0.78	0.53	166M	1	0.3 [74]
	Diff.	ADM [26]	10.94	101.0	0.69	0.63	554M	250	168 [74]
	Diff.	CDM [36]	4.88	158.7	—	—	—	8100	—
	Diff.	LDM-4-G [70]	3.60	247.7	—	—	400M	250	—
	Diff.	DiT-L/2 [63]	5.02	167.2	0.75	0.57	458M	250	31
	Diff.	DiT-XL/2 [63]	2.27	278.2	0.83	0.57	675M	250	45
	Diff.	L-DiT-3B [3]	2.10	304.4	0.82	0.60	3.0B	250	>45
	Diff.	L-DiT-7B [3]	2.28	316.2	0.83	0.58	7.0B	250	>45
	Mask.	MaskGIT [17]	6.18	182.1	0.80	0.51	227M	8	0.5 [17]
	Mask.	RCG (cond.) [51]	3.49	215.5	—	—	502M	20	1.9 [51]
	AR	VQVAE-2† [68]	31.11	~45	0.36	0.57	13.5B	5120	—
VAR	AR	VQGAN† [30]	18.65	80.4	0.78	0.26	227M	256	19 [17]
	AR	VQGAN [30]	15.78	74.3	—	—	1.4B	256	24
	AR	VQGAN-re [30]	5.20	280.3	—	—	1.4B	256	24
	AR	ViTVQ [92]	4.17	175.1	—	—	1.7B	1024	>24
	AR	ViTVQ-re [92]	3.04	227.4	—	—	1.7B	1024	>24
	AR	RQTran. [50]	7.55	134.0	—	—	3.8B	68	21
	AR	RQTran.-re [50]	3.80	323.7	—	—	3.8B	68	21
	VAR	VAR- <i>d</i> 16	3.30	274.4	0.84	0.51	310M	10	0.4
	VAR	VAR- <i>d</i> 20	2.57	302.6	0.83	0.56	600M	10	0.5
	VAR	VAR- <i>d</i> 24	2.09	312.9	0.82	0.59	1.0B	10	0.6
	VAR	VAR- <i>d</i> 30	1.92	323.1	0.82	0.59	2.0B	10	1
	VAR	VAR- <i>d</i> 30-re	1.73	350.2	0.82	0.60	2.0B	10	1
(validation data)			1.78	250.9	0.75	0.67			

VAR is comparable or even **outperforms** diffusion models in class-conditional generation.

Fast Inference of Visual AutoRegressive Model (VAR)

Table 1: **Generative model family comparison on class-conditional ImageNet 256×256 .** “↓” or “↑” indicate lower or higher values are better. Metrics include Fréchet inception distance (FID), inception score (IS), precision (Pre) and recall (rec). “#Step”: the number of model runs needed to generate an image. Wall-clock inference time relative to VAR is reported. Models with the suffix “-re” used rejection sampling. †: taken from MaskGIT [17].

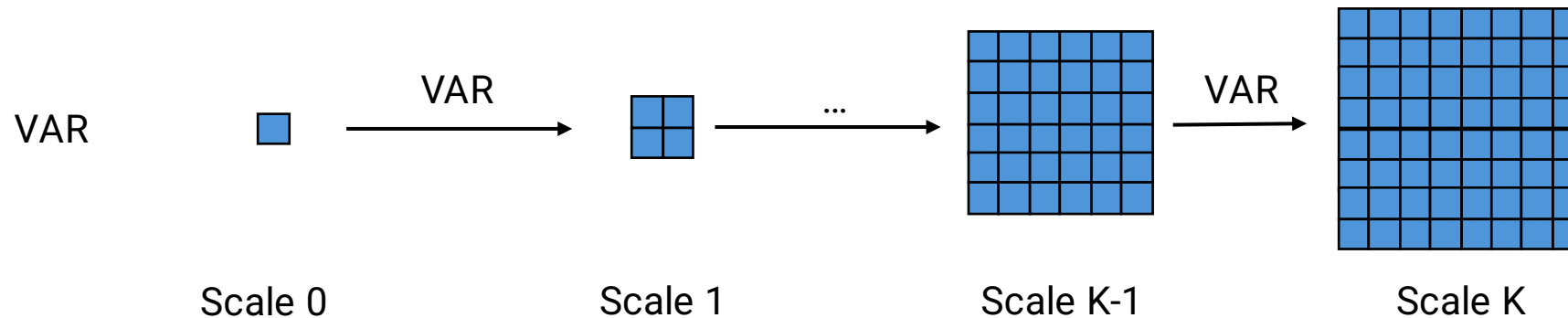
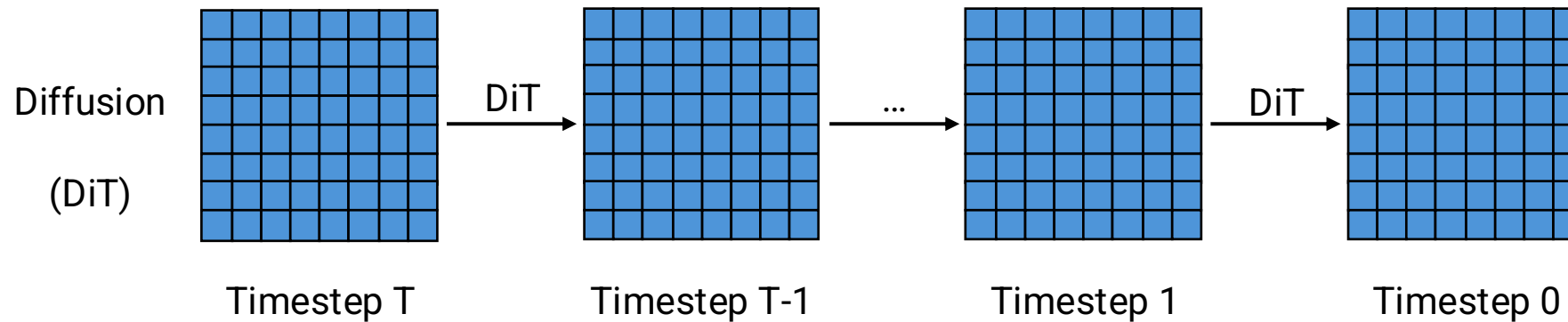
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	(validation data)	1.78	236.9	0.75	0.67			

VAR has **faster inference time** than diffusion models and previous AR models.

Why is VAR fast?

Why is VAR's Inference Faster than Diffusion's?

In latent space,



VAR is **faster** than diffusion models at inference, because

1. **Fewer** sequential **forward** passes

T (50~250) $\gg$ K (10~13)

2. **Less high-resolution processing**

Diffusion: fixed full latent at every step

VAR: small $\rightarrow$ large scales

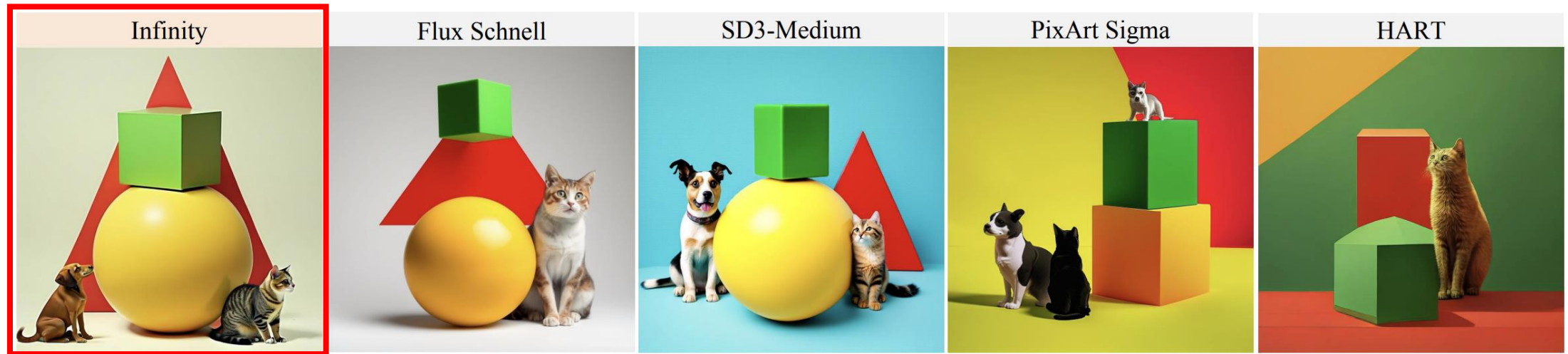
Result of Visual AutoRegressive Model (VAR) – T2I

Table 1: Evaluation on the GenEval [24] and DPG [29] benchmark. † result is with prompt rewriting.

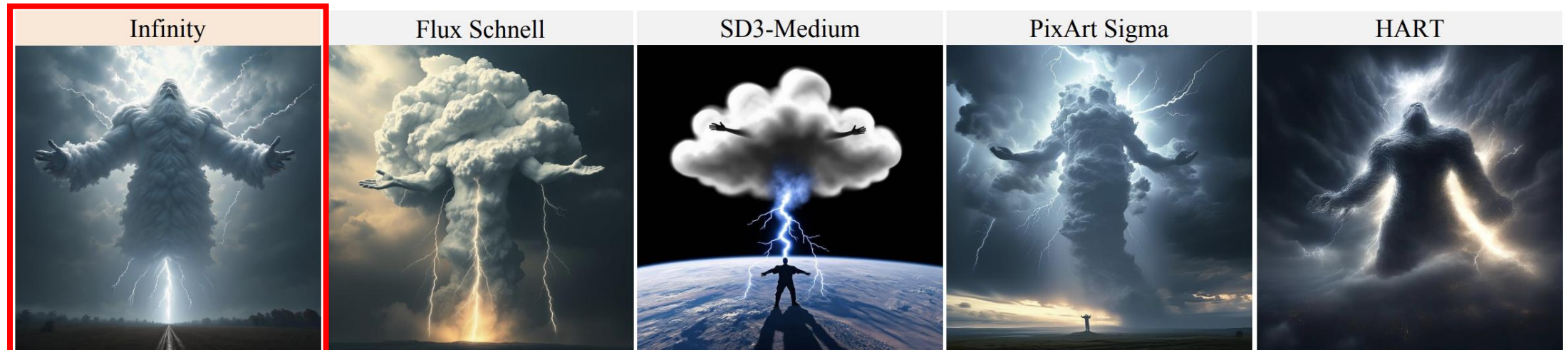
Methods	# Params	GenEval↑				DPG↑		
		Two Obj.	Position	Color Attri.	Overall	Global	Relation	Overall
Diffusion Models								
LDM [49]	1.4B	0.29	0.02	0.05	0.37	-	-	-
SDv1.5 [49]	0.9B	0.38	0.04	0.06	0.43	74.63	73.49	63.18
PixArt-alpha [13]	0.6B	0.50	0.08	0.07	0.48	74.97	82.57	71.11
SDv2.1 [49]	0.9B	0.51	0.07	0.17	0.50	77.67	80.72	68.09
DALL-E 2 [45]	6.5B	0.66	0.10	0.19	0.52	-	-	-
DALL-E 3 [7]	-	-	-	-	0.67 [†]	90.97	90.58	83.50
SDXL [43]	2.6B	0.74	0.15	0.23	0.55	83.27	86.76	74.65
PixArt-Sigma [12]	0.6B	0.62	0.14	0.27	0.55	86.89	86.59	80.54
SD3 (d=24) [21]	2B	0.74	0.34	0.36	0.62	-	-	84.08
SD3 (d=38) [21]	8B	0.89	0.34	0.47	0.71	-	-	-
AutoRegressive Models								
LlamaGen [55]	0.8B	0.34	0.07	0.04	0.32			65.16
Chameleon [59]	7B	-	-	-	0.39	-	-	-
HART [58]	732M	-	-	-	0.56	-	-	80.89
Show-o [72]	1.3B	0.80	0.31	0.50	0.68	-	-	67.48
Emu3 [67]	8.5B	0.81 [†]	0.49 [†]	0.45 [†]	0.66 [†]	-	-	81.60
Infinity	2B	0.85 [†]	0.49 [†]	0.57 [†]	0.73 [†]	93.11	90.76	83.46

VAR (Infinity) demonstrates **competitive performance** with diffusion models in **T2I** generation.

Result of Visual AutoRegressive Model (VAR) – T2I

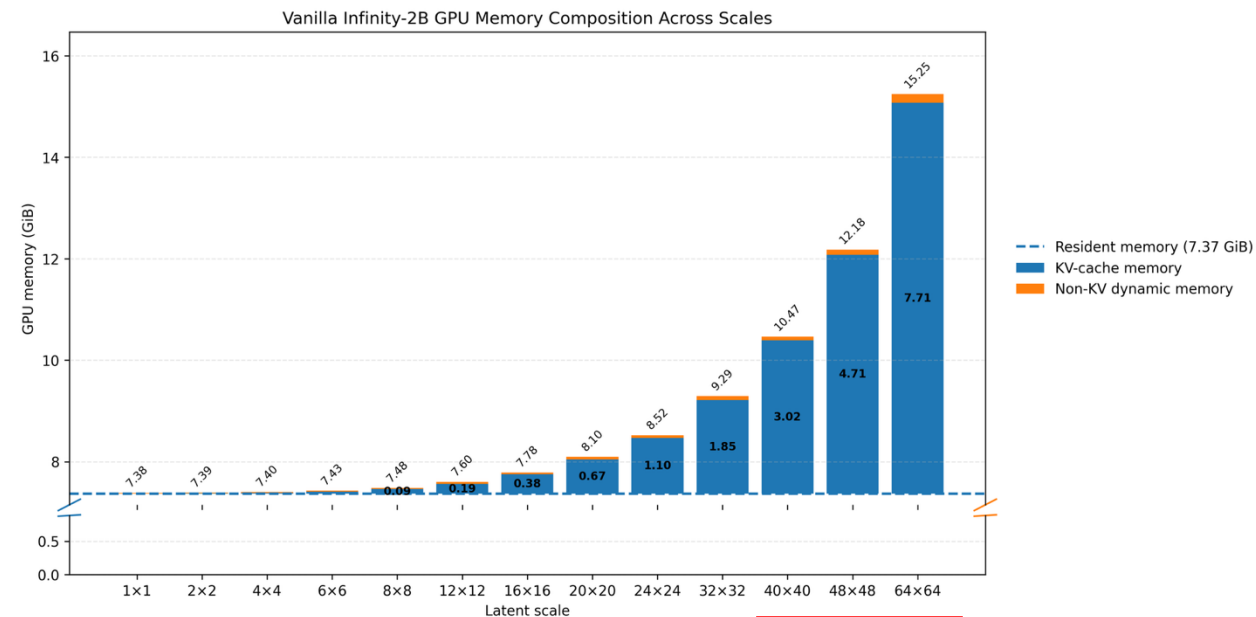
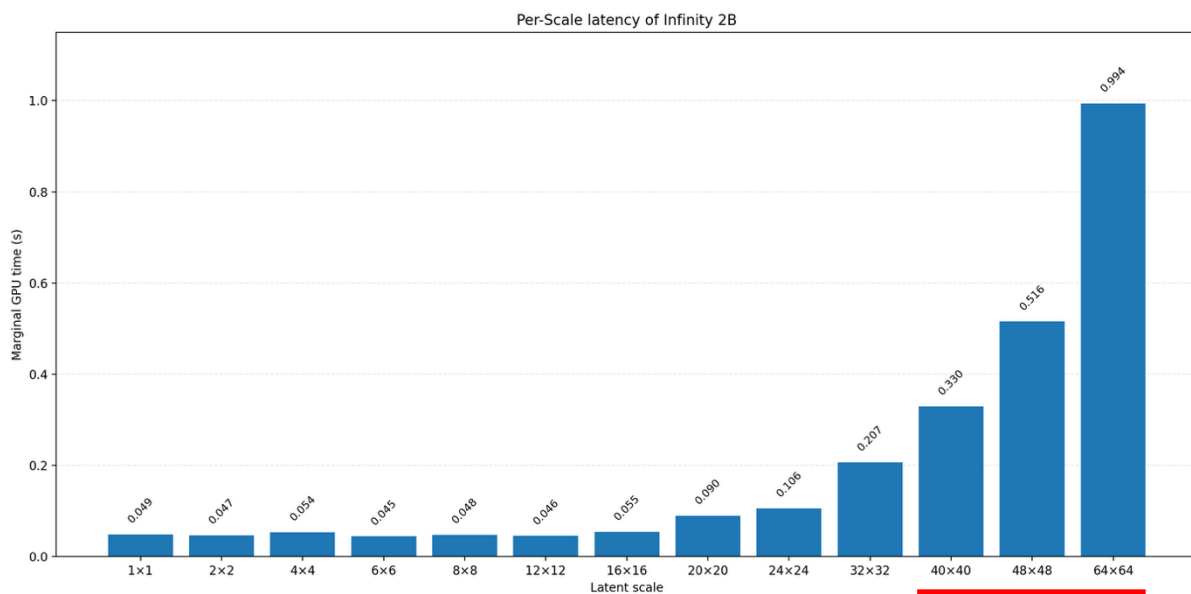


a green cube on the top of a yellow sphere, the behind is a red triangle. A cat is on the right and a dog is on the left



a giant, towering cloud in the shape of a man looms over the earth, with arms outstretched like a mythical deity. The cloud man, composed of swirling, gray and white vapors, shoots lightning bolts down to the earth, piercing through the dark sky. The bolts illuminate the scene with a dramatic, otherworldly light, casting sharp shadows on the ground below.

Bottlenecks of VAR Inference



Low costs at early scales(steps), **but high costs at later scales(steps)!**

Previous Works on Efficient VAR

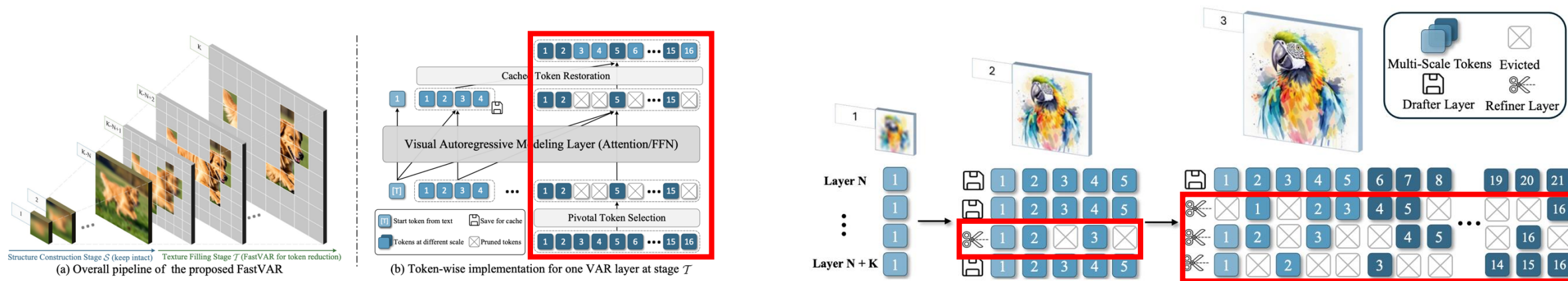


Figure 4. (a) We keep the **Structure Construction Stage** as standard VAR and additionally store token maps at the $(K - N)$ -th step as the cache. Then the **Texture Filling Stage** applies FastVAR to process only pivotal tokens for fast inference. (b) For Attention or FFN layers using FastVAR, we first employ the Pivotal Token Selection (PTS) for token reduction. After the model forward, we develop the Cached Token Restoration (CTR) to restore the original token numbers using the caches from the previous scale step.

Figure 4: Overview of ScaleKV. Our method categorizes transformer layers into drafters (require extensive cache for global context) or refiners (process local details with minimal cache). This scale-wise identification enables adaptive cache allocation based on each layer's computational demands.

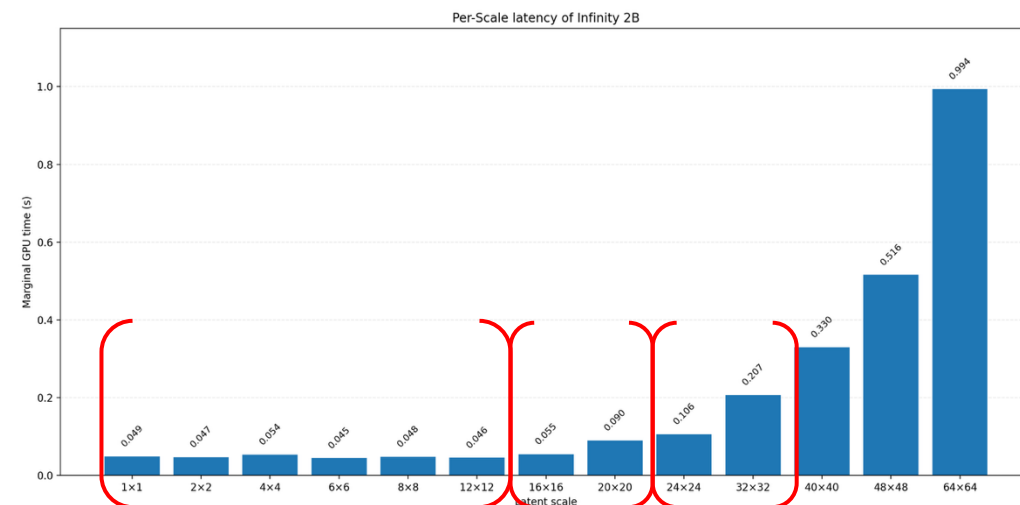
How to deal with later step bottleneck?

1. **Token pruning:** Process only important visual tokens
2. **KV cache compression:** Retain only important historical K/V states

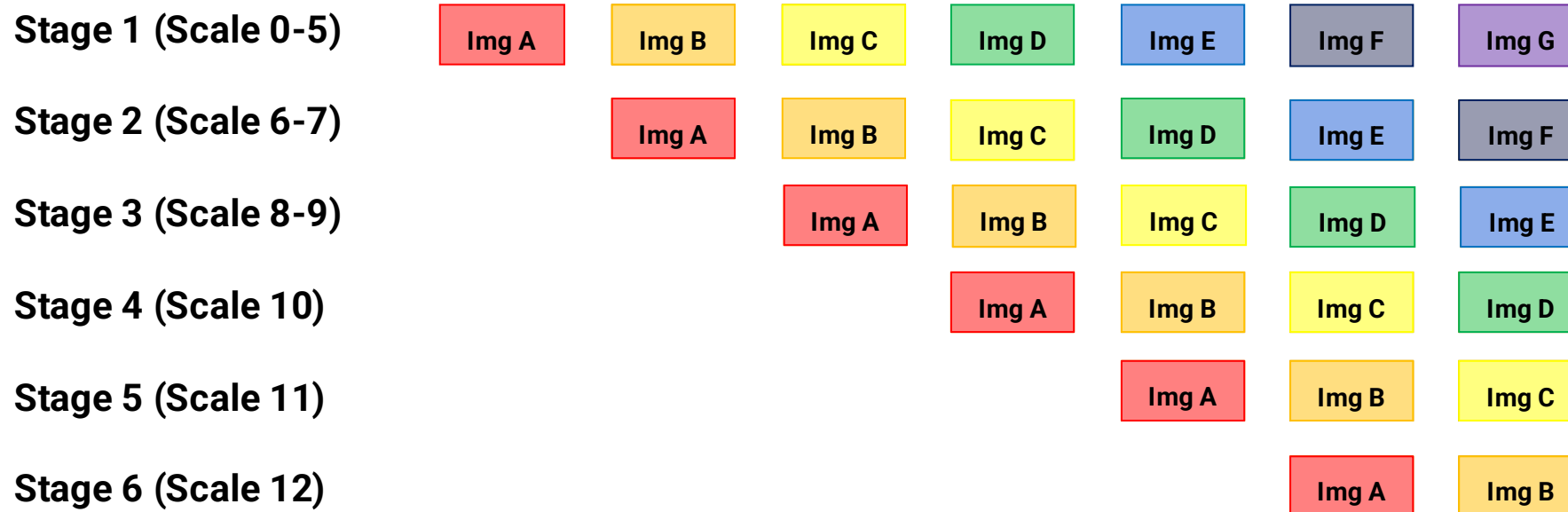
Our Research Direction 1 – Pipelining

Multi Sample VAR Pipeline

Early Scales(steps) are less time consuming while later scales(steps) are more time consuming.

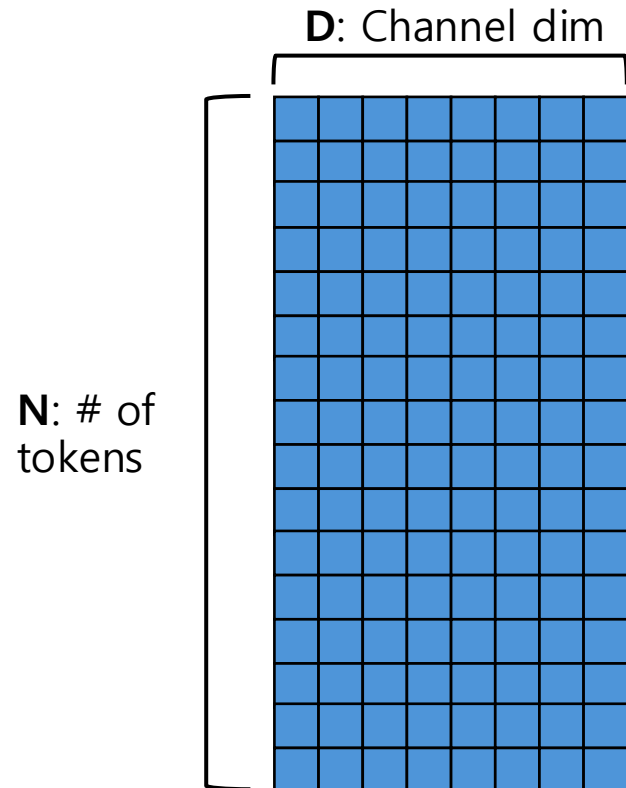


Time



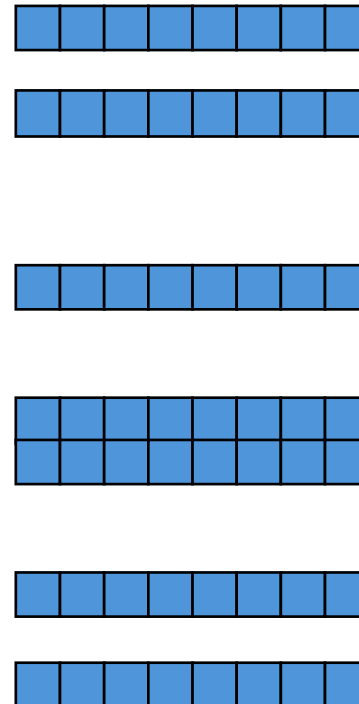
Our Research Direction 2 – KV Cache Channel Pruning

Full KV Cache



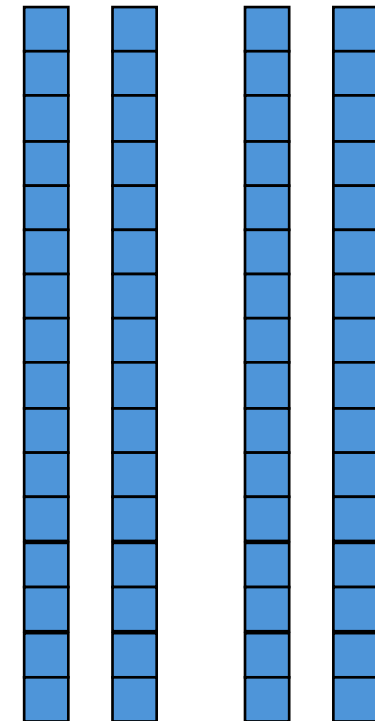
Previous KV Cache Compression in VAR

- **Compress** in **token** axis



Our Direction

- **Compress** in **channel** axis



Conclusions

1. **VAR** enables **efficient** image generation via **next-scale prediction**

- Predicts tokens **in parallel within each scale**
- Requires **fewer sequential forward** passes than conventional AR / diffusion baselines

2. **Efficiency** is still **limited** by **later scales(steps)**

- **Computation** and **KV-cache memory** become heavily concentrated at high-resolution scales

3. **Our research** focuses on the **remaining VAR bottleneck**

- Multi-sample pipelining for better throughput
- KV-cache channel compression for lower memory cost

Thank you!

Appendix: VAR Inference Pseudo Code

Algorithm 2 Vanilla VAR Inference

Require: Scale resolutions $\{(h_k, w_k)\}_{k=1}^K$; VAR model ϕ ; codebook $\mathcal{Z} \in \mathbb{R}^{V \times c}$; token embedding $\psi : \mathbb{R}^c \rightarrow \mathbb{R}^d$; image decoder $\mathcal{D}$.

Ensure: Generated image $I \in \mathbb{R}^{B \times 3 \times H_I \times W_I}$.

```

1:  $H \leftarrow h_K, W \leftarrow w_K$ 
2:  $F_0 \leftarrow \mathbf{0} \in \mathbb{R}^{B \times c \times H \times W}$ 
3: for  $k = 1, \dots, K$  do
4:    $N_k \leftarrow h_k w_k$ 
5:   if  $k = 1$  then
6:      $\tilde{F}_0 \leftarrow \langle \text{SOS} \rangle \in \mathbb{R}^{B \times 1 \times d}$ 
7:   else
8:      $G_{k-1}^{(k)} \leftarrow \text{Down}(F_{k-1}, (h_k, w_k)) \in \mathbb{R}^{B \times c \times h_k \times w_k}$ 
9:      $\tilde{F}_{k-1} \leftarrow \psi(\text{Flatten}(G_{k-1}^{(k)})) \in \mathbb{R}^{B \times N_k \times d}$ 
10:  end if
11:   $L_k \leftarrow \phi(\tilde{F}_{k-1}) \in \mathbb{R}^{B \times N_k \times V}$   $\triangleright$  next-scale prediction
12:   $z_k \leftarrow \text{Sample}(L_k) \in \{1, \dots, V\}^{B \times N_k}$   $\triangleright$  token sampling
13:   $E_k \leftarrow \mathcal{Z}[z_k] \in \mathbb{R}^{B \times N_k \times c}$   $\triangleright$  codebook lookup
14:   $R_k \leftarrow \text{Reshape}(E_k, (c, h_k, w_k)) \in \mathbb{R}^{B \times c \times h_k \times w_k}$   $\triangleright$  residual map
15:   $F_k \leftarrow F_{k-1} + \text{Up}(R_k, (H, W)) \in \mathbb{R}^{B \times c \times H \times W}$   $\triangleright$  residual accumulation
16: end for
17:  $I \leftarrow \mathcal{D}(F_K) \in \mathbb{R}^{B \times 3 \times H_I \times W_I}$ 
18: return  $I$ 

```

Algorithm 3 Attention Computation inside VAR Model at Scale k

Require: Current-scale tokens $\mathbf{X}_k^0 = \tilde{\mathbf{F}}_{k-1} \in \mathbb{R}^{B \times N_k \times d}$; cached keys and values $\{\mathbf{K}_{<k}^{\ell,h}, \mathbf{V}_{<k}^{\ell,h}\}$; number of Transformer layers L ; number of attention heads H_a

Ensure: Next-scale logits $\mathbf{L}_k \in \mathbb{R}^{B \times N_k \times V}$

```

1: for  $\ell = 1, \dots, L$  do
2:    $\bar{\mathbf{X}}_k^\ell \leftarrow \text{Norm}_\ell(\mathbf{X}_k^{\ell-1})$ 
3:   for  $h = 1, \dots, H_a$  do
4:      $\mathbf{Q}_k^{\ell,h} \leftarrow \bar{\mathbf{X}}_k^\ell \mathbf{W}_Q^{\ell,h} \in \mathbb{R}^{B \times N_k \times d_h}$ 
5:      $\mathbf{K}_k^{\ell,h} \leftarrow \bar{\mathbf{X}}_k^\ell \mathbf{W}_K^{\ell,h} \in \mathbb{R}^{B \times N_k \times d_h}$ 
6:      $\mathbf{V}_k^{\ell,h} \leftarrow \bar{\mathbf{X}}_k^\ell \mathbf{W}_V^{\ell,h} \in \mathbb{R}^{B \times N_k \times d_h}$ 
7:      $\mathbf{K}_{\leq k}^{\ell,h} \leftarrow \text{Concat}(\mathbf{K}_{<k}^{\ell,h}, \mathbf{K}_k^{\ell,h}) \in \mathbb{R}^{B \times T_k \times d_h}$ 
8:      $\mathbf{V}_{\leq k}^{\ell,h} \leftarrow \text{Concat}(\mathbf{V}_{<k}^{\ell,h}, \mathbf{V}_k^{\ell,h}) \in \mathbb{R}^{B \times T_k \times d_h}$ 
9:      $\mathbf{S}_k^{\ell,h} \leftarrow \frac{\mathbf{Q}_k^{\ell,h} (\mathbf{K}_{\leq k}^{\ell,h})^\top}{\sqrt{d_h}} \in \mathbb{R}^{B \times N_k \times T_k}$ 
10:     $\mathbf{A}_k^{\ell,h} \leftarrow \text{Softmax}_{\text{key}}(\mathbf{S}_k^{\ell,h}) \in \mathbb{R}^{B \times N_k \times T_k}$ 
11:     $\mathbf{O}_k^{\ell,h} \leftarrow \mathbf{A}_k^{\ell,h} \mathbf{V}_{\leq k}^{\ell,h} \in \mathbb{R}^{B \times N_k \times d_h}$ 
12:  end for
13:   $\mathbf{O}_k^\ell \leftarrow \text{Concat}(\mathbf{O}_k^{\ell,1}, \dots, \mathbf{O}_k^{\ell,H_a}) \mathbf{W}_O^\ell$ 
14:   $\mathbf{Y}_k^\ell \leftarrow \mathbf{X}_k^{\ell-1} + \mathbf{O}_k^\ell$ 
15:   $\mathbf{X}_k^\ell \leftarrow \mathbf{Y}_k^\ell + \text{FFN}_\ell(\text{Norm}_\ell^{\text{FFN}}(\mathbf{Y}_k^\ell))$ 
16:   $\mathbf{K}_{<k+1}^{\ell,h} \leftarrow \mathbf{K}_{\leq k}^{\ell,h}, \mathbf{V}_{<k+1}^{\ell,h} \leftarrow \mathbf{V}_{\leq k}^{\ell,h}$ 
17: end for
18:  $\mathbf{L}_k \leftarrow \text{PredictionHead}(\mathbf{X}_k^L)$ 
19: return  $\mathbf{L}_k$ 

```

$$T_k = \sum_{s=1}^k N_s.$$